Alex I.

SOFTWARE ENGINEER

Education
Computer Science and Software Engineering

Language proficiency
English – B2
Polish – B1
Swedish – A2

Domains
Telecommunications
Networks
System Administration
UAV Systems

Software engineer with 8+ years of commercial experience.
I possess a strong background in both hardware and software development. My expertise mostly revolves around C/C++ software development, where I appreciate the complexity and versatility of the language, and MATLAB, where I have extensive experience in developing algorithms and applications for image analysis. I am eager to work on projects that prioritize performance and innovative features and enthusiastic about collaborating on challenging projects that allow me to leverage the full capabilities of these technologies and contribute to cutting-edge solutions.

Programming Languages
C++, C, C#, Java, JavaScript, MATLAB, Scala, HTML.

Operating Systems
Windows, Windows Server, Embedded Linux.

Tools
Wireshark, Yocto Project, tcpdump, CMake, qmake, gcc, GDB, Valgrind, Visual Studio, Resharper C++.

Frameworks & Libraries
GStreamer, Qt, Qt Widgets, QSS, .NET, Node.js, WinForms, OpenSSL, PCL, V4L2, jQuery, Open3D.

Multimedia
WebRTC, Linphone-SDK, liblinphone, mediasoup, libmediasoupclient.

Databases
MySQL, PostgreSQL, SQLite, Microsoft SQL Server.

Network
SDP, TCP/IP, Nginx, SIP, SNMP, DHCP, TFTP.

Cloud

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AWS ().

Embedded
BitBake, NXP i.MX.

Source control systems
Git, GitLab, Bitbucket.

Other
FTP, RTP/RTCP, STUN/TURN/ICE, TLS/SRTP, QoS/DSCP, VLAN, Windows IoT.

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Projects

SIP DESK PHONE PLATFORM FOR AUDIO AND VIDEO COMMUNICATION

A SIP desk phone platform for audio and video communication, supporting both camera-enabled and audio-only device models. The project included the native phone application, web management interface, automated provisioning tools, embedded system customization, and security module refactoring. It covered everyday calling workflows such as SIP registration, incoming and outgoing calls, video calls, call hold, transfer, conferencing, phonebook access, direct IP dialing, remote configuration, and device setup.

Project roles
Software Engineer

Period
11.2023 – till now

Responsibilities

- Decoupled cryptographic logic from Linphone-SDK into a generic base wrapper, enabling integration with external custom and hardware security modules;
- Performed security analysis and backported CVE fixes and OpenSSL migrations to legacy library branches;
- Refactored SIP call handling toward MVC/MVVM by separating UI rendering from call state, media negotiation, and Linphone-SDK integration;
- Decomposed legacy Qt Widgets call-flow code into domain, presentation, and SDK boundaries while preserving QObject signal compatibility;
- Hardened video-call and conference UI for edge cases like held calls, second lines, remote hangups, stale overlays, and video stream switching;
- Extended Linphone-SDK backend logic to implement SIP failover, multi-homed IP routing, and backup-server connectivity for high availability;
- Optimized RTP/RTCP packet flows, NAT traversal (STUN/TURN/ICE), and QoS (DSCP/CoS) tagging for low-latency audio/video traffic;
- Developed WebRTC video-conferencing UI and signaling flows, including room management, participant rosters, tile layouts, and media controls;
- Expanded WinForms provisioning utility on .NET 8 into an all-in-one server suite with integrated DHCP, FTP provisioning, and local SIP services;
- Automated Windows IoT server network setup, project file formats, and zero-touch device workflows using SNMP, DHCP options, FTP, and SQLite;

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- Troubleshot low-level network issues, SDP negotiation, TLS/SRTP encryption, and A/V sync using Wireshark and tcpdump on Linux embedded targets;
- Customised Yocto/BitBake layers for NXP i.MX targets and configured target network services (systemd-networkd, VLAN tagging).

Environment
C++17, C#, JavaScript, Embedded Linux, Wireshark, Yocto, tcpdump, CMake, qmake, GCC, gdb, Valgrind, GStreamer, Qt 5.15.7, Qt Widgets, QSS, .NET 8, Node.js, WinForms, OpenSSL, V4L2, WebRTC, Linphone-SDK, liblinphone, mediasoup, libmediasoupclient, SQLite, SDP, TCP/IP, SIP, SNMP, DHCP, TFTP, BitBake, NXP i.MX, Git, GitLab, FTP, RTP/RTCP, STUN/TURN/ICE, TLS/SRTP, QoS/DSCP, VLAN, Windows IoT.

LIBRARY SYSTEM

A robust and highly scalable library management system with over 40 years of operational history, serving a diverse client base across the globe. The system is built using a combination of C, Java, and Scala, leveraging the strengths of each language to deliver a reliable, efficient, and feature-rich platform. Designed to handle high volumes of data and complex workflows, it has continually evolved to meet the needs of modern libraries while maintaining compatibility with legacy components.

Project roles
Software Engineer

Period
04.2021 – 11.2023

Responsibilities
- Optimized and maintained the existing codebase by resolving complex bugs and implementing performance enhancements;
- Developed new backend and frontend features for socket-based communication;
- Led development efforts as a key contributor in C language, driving the successful implementation of critical features and solutions;
- Refactored existing Scala API code to improve maintainability and scalability;
- Worked with AWS pipelines, automating CI/CD processes for smoother deployments and reducing development cycle times;
- Backported CVE fixes from the upstream OpenSSL into the local fork resolving conflicts;
- Modified database tables to support new feature

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implementations, improving schema design to align with updated system workflows;
- Estimated tasks with precision, ensuring realistic timelines and efficient resource allocation for development milestones;
- Conducted thorough code reviews, ensuring code quality, adherence to standards, and knowledge sharing within the basic team and client's team;
- Refactored legacy application code to improve maintainability, scalability, and overall system efficiency.

Environment

C, Java 11, Scala, Windows Server, gdb, OpenSSL, MySQL, PostgreSQL, Amazon Web Services, Git, BitBucket.

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CORPORATE SYSTEM FOR THE OIL AND GAS INDUSTRY

Corporate system tailored for the oil and gas industry, designed for seamless internal use across various devices. The system leverages C++17 for robust backend development and JavaScript, HTML, and jQuery for interactive frontends. With Microsoft SQL Server handling data storage and Nginx ensuring smooth server operations, it offers a reliable and high-performance solution.

Development processes were streamlined with Resharper C++ for enhanced code quality, CMake for efficient build automation, and version control tools like Git and BitBucket. The solution was optimized for deployment in Windows environments, ensuring compatibility and ease of use across devices.

Project roles
Software Engineer

Period
08.2020 – 04.2021

Responsibilities

- Engineered efficient back-end software solutions, ensuring optimal performance for critical operations in the oil and gas industry;
- Contributed as a core developer, implementing key functionalities and supporting project milestones;
- Provided accurate task estimations, enabling effective planning and resource management for project deliverables;
- Maintained and enhanced the existing codebase by resolving bugs, optimizing performance, and reducing technical debt;
- Removed outdated legacy components, modernizing the system to improve reliability and maintainability.

Environment
C++17, JavaScript, HTML, Windows, CMake, Resharper C++, jQuery, MySQL, Microsoft SQL Server, Nginx, Git, BitBucket.

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3D RECONSTRUCTION SOFTWARE FOR INDUSTRIAL APPLICATIONS

Developed cutting-edge software aimed at generating precise 3D models by processing data captured by multiple 2D laser sensors. The solution is tailored for industries requiring highly accurate 3D spatial data, such as manufacturing, robotics, and quality control.

Project roles
Software Engineer

Period
03.2020 – 08.2020

Responsibilities

- Developed software for generating 3D models from data collected by multiple 2D laser sensors, significantly enhancing accuracy in 3D reconstruction;
- Utilized the Point Cloud Library (PCL) for efficient processing and manipulation of point cloud data, optimizing data handling and performance;
- Integrated Open3D to facilitate advanced visualization and analysis of 3D models, improving user interaction and understanding of spatial data;
- Employed Visual Studio for coding, debugging, and optimizing performance, ensuring a robust and efficient software solution.

Environment
C++, Visual Studio, PCL, Open3D, Git.

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SOFTWARE MODULES FOR UAV

Developed innovative software solutions and algorithms to enhance the performance of optical sensors used in satellites and UAVs. The project included advanced photo and video processing, data analysis, and testing to improve system reliability, accuracy, and image quality for remote sensing applications.

Project roles
Software Engineer

Period
08.2018 – 12.2019

Responsibilities

- Developed software modules for digital image processing systems used in Unmanned Aerial Vehicles;
- Implemented a debayer algorithm to convert raw sensor data into full-color images, significantly improving image quality and accuracy for better visual data interpretation;
- Created image format conversion tools to support multiple formats, enhancing compatibility and integration across different systems and platforms;
- Developed an image stabilization algorithm to reduce motion-induced distortions, improving video clarity in real-time UAV operations for clearer footage capture;
- Designed an image fusion algorithm to combine data from multiple sensors, increasing detail and contrast for improved target recognition and analysis.

Environment
MATLAB, PostgreSQL, Git, BitBucket.

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Professional skills

SKILLS		EXPERIENCE IN YEARS	LAST USED
Programming Languages	C++	4	2026
	C	3	2023
	C#	3	2026
	Java	3	2023
	JavaScript	3	2026
	MATLAB	1	2019
	Scala	3	2023
	HTML	1	2021
Operating Systems	Windows	1	2021
	Windows Server	3	2023
	Embedded Linux	3	2026
Tools	Wireshark	3	2026
	Yocto Project	3	2026
	tcpdump	3	2026
	CMake	3	2026
	qmake	3	2026
	gcc	3	2026
	GDB	5	2026
	Valgrind	3	2026
	Visual Studio	1	2020
	Resharper C++	1	2021
Frameworks & Libraries	GStreamer	3	2026
	Qt	3	2026
	Qt Widgets	3	2026
	QSS	3	2026

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SKILLS		EXPERIENCE IN YEARS	LAST USED
	.NET	3	2026
	Node.js	3	2026
	WinForms	3	2026
	OpenSSL	5	2026
	PCL	1	2020
	V4L2	3	2026
	jQuery	1	2021
	Open3D	1	2020
Multimedia	WebRTC	3	2026
	Linphone-SDK	3	2026
	liblinphone	3	2026
	mediasoup	3	2026
	libmediasoupclient	3	2026
Databases	MySQL	3	2023
	PostgreSQL	4	2023
	SQLite	3	2026
	Microsoft SQL Server	1	2021
Network	SDP	3	2026
	TCP/IP	3	2026
	Nginx	1	2021
	SIP	3	2026
	SNMP	3	2026
	DHCP	3	2026
	TFTP	3	2026
Cloud	AWS	3	2023

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SKILLS		EXPERIENCE IN YEARS	LAST USED
Embedded	BitBake	3	2026
	NXP i.MX	3	2026
Source control systems	Git	8	2026
	GitLab	3	2026
	Bitbucket	5	2023
Other	FTP	3	2026
	RTP/RTCP	3	2026
	STUN/TURN/ICE	3	2026
	TLS/SRTP	3	2026
	QoS/DSCP	3	2026
	VLAN	3	2026
	Windows IoT	3	2026

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